

“WORKING ONLY ON ARDUINO SYSTEMS Does Not Make Someone AN EMBEDDED SYSTEMS ENGINEER”

What if every engineering student could build products that power the next generation of smart devices? Jayakumar Balasubramanian and Maaz Jukaku from Emertxe shared how hands-on learning and real-world systems can turn learners into industry-ready innovators in a recent conversation with Rahul Chopra and Nidhi Agarwal from Electronics For You.

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What is an embedded systems engineer? For instance, if I build Arduino projects, does that make me one, or am I still just a software developer?

An embedded system combines customised hardware and software to perform a specific function or a defined set of functions. Anyone who designs and builds such systems is considered an embedded systems engineer, though expertise ranges from simple systems such as timers to advanced systems used in electric vehicles (EVs) or smart cities. **Arduino makes it easy to begin learning how hardware and software work together, but working only on Arduino systems does not make someone an embedded systems engineer.** It serves as an introduction to the core concepts.

To become a professional, growth comes from learning bare-metal programming on microcontrollers such as STM32 or ARM, using languages such as C or C++, understanding communication protocols, and working on domain-specific applications including automotive, medical, or industrial systems. As

learners advance, they can work with real-time operating systems, such as FreeRTOS or embedded Linux, which are used in systems such as car dashboards and automation devices. Strong fundamentals in problem-solving, algorithms, optimisation, and programming are essential. Start small, build the foundation, move to advanced tools and operating systems, and eventually specialise in a field such as IoT, robotics, or automotive to become a professional embedded systems engineer.

How should I train for embedded systems?

To excel in embedded systems, strong problem-solving skills, algorithmic thinking, and solid fundamentals are essential. It is similar to driving a car: the model does not matter as much as knowing the basics, such as the accelerator, brake, and clutch. Similarly, boards and tools are enablers; what counts is understanding the core concepts. Focus on algorithms, optimisation, programming in C, bare-metal microcontroller coding, and embedded Linux. Take a top-down approach: decide

which product to build and learn the skills required. Explore fields such as automotive, consumer electronics, healthcare, or logistics. Building and showcasing a working solution helps the learner become industry-ready.

Is embedded systems engineering mostly about software and less about electronic circuitry?

Yes, today it is software-focused primarily because smart features in devices are software-driven. However, knowing the basics of hardware remains important. Engineers do not need to design circuits or chips, yet they must understand how hardware works, read datasheets, and interface with parts such as sensors, motors, and displays. For instance, when building a robotic arm, the engineer may not design the board, but must program the controller, communicate with servo motors, and interpret their signals.

In short, an embedded engineer needs strong software skills and sufficient hardware knowledge to make both work together. It begins with learning to read and apply datasheets and using that understanding to build functional systems.